

WELL COMPLETION REPORT (SYNTHETIC)

OIL-NHK-009 - Nahorkatiya-009

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000098
Field / District	Nahorkatiya / Dibrugarh
Block	PEL-ASSAM-02
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	2023-10-06 / 2023-12-23
Measured / TVD	2536 m / 2517 m
Trajectory	Directional; inc 33.4 deg; az 1.7 deg
Formation Interval	1190-2007 m
Lithology	Fine-medium sandstone with shale interbeds
Pore Pressure Reference	2604 psi
Fracture Gradient	14.98 ppg
Mud System	Glycol-PHPA WBM
NWIS Well Risk	Moderate (54.8/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	634	Surface
9 5/8 in casing	1470	Intermediate
7 in liner/casing	2231	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2023-11-22 1615 m	Kick	High	Flow-out exceeded flow-in and gas units increased. Mitigation: Re-establish overbalance and monitor pit/flow balance. Lesson: Treat pit gain plus gas increase as an early influx indicator.
2023-11-25 1703 m	Torque Spike	High	Torque rose without corresponding pressure anomaly. Mitigation: Reduce WOB/RPM and circulate before resuming normal parameters. Lesson: Use offset torque trend to set adaptive operating envelope.
2023-11-27 1762 m	Kick	High	Unexpected background gas rise preceded influx. Mitigation: Maintain trip margin and use conservative connection practices. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
2023-12-02 1922 m	Stuck Pipe	High	Rotary movement deteriorated progressively. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2023-12-03 1936 m	Pressure Spike	High	Circulating pressure climbed without equivalent torque change. Mitigation: Reduce pump rate, verify restriction and restore circulation gradually. Lesson: Correlate pressure anomalies with flow before escalating risk.
2023-10-31 938 m	Torque Spike	Medium	Rotary speed fluctuated during constant pump rate. Mitigation: Reduce WOB/RPM and circulate before resuming normal parameters. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2023-12-15 2295 m	Stuck Pipe	Medium	High drag was recorded while circulating remained possible. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency.

Completion Summary

The synthetic well reached 2536 m MD in the Tipam interval. The NWIS knowledge record links 10 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.